Professional development formats and assessment self-efficacy among lower-secondary teachers

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Purpose

This project examines whether the format of professional development (PD), independent of its content, is associated with lower-secondary teachers' self-efficacy in using assessment strategies. Data are from the TALIS 2024 survey, restricted to Netherlands lower-secondary (ISCED 2) teachers (n = 1,271).

The practical motivation behind this research project is that systems routinely make PD format decisions, workshops, conferences, peer networks, without strong evidence that format independently shapes teacher confidence. Most research on PD effectiveness asks *what* teachers learn, not *how* the learning is structured. The Netherlands is a useful test case since the survey shows wide variation in which PD formats teachers use.

Data

Source: OECD Teaching and Learning International Survey (TALIS) 2024, international teacher file (`ttgintt4`)

Sample: Netherlands lower-secondary teachers (`IDCNTPOP = NLD2`): **n = 1,271**, 630 variables (1,179 with a valid outcome). Dutch primary (ISCED 1) teachers in the file are excluded.

PD items are coded 1/2/3 = Yes (in person / online / both) and 4 = No; Yes is pooled against No. `TT4G20A` *(courses/workshops) has no valid Netherlands responses.*

Key variables:

Role	Variable	Description
Outcome	`TT4G27J`	Self-efficacy: "Use a variety of assessment strategies" (1–4)
Predictor	`TT4G20A`	PD format: Courses / workshops / seminars
Predictor	`TT4G20B`	PD format: Education conferences
Predictor	`TT4G20C`	PD format: Formal qualification programme
Predictor	`TT4G20D`	PD format: Visits to other schools
Predictor	`TT4G20E`	PD format: Visits to business premises
Predictor	`TT4G20F`	PD format: Lesson observation and reflection
Predictor	`TT4G20G`	PD format: Coaching / mentoring
Predictor	`TT4G20H`	PD format: Teacher networks
Predictor	`TT4G20I`	PD format: Self-initiated learning
Not analysed	`TT4G21M`	Whether PD covered student assessment analysis: flagged for future work

Methods

1. **Data ingestion**: Chunked CSV reading to filter Netherlands lower-secondary rows from the ~278k-row international file (too large to load into memory at once)
2. **Sentinel recode**: TALIS uses numeric codes for missing or inapplicable responses (8, 9, 96–99, 996–999); these were recoded to `NaN` on a clean copy of the data, preserving the original
3. **Variable labelling**: SPSS metadata extracted via `pyreadstat` to map cryptic column codes (e.g., `TT4G27J`) to full question text
4. **Descriptive analysis**: PD participation rates per format; outcome distribution including mean and SD
5. **Inferential analysis**: Mann-Whitney U tests comparing assessment self-efficacy between PD participants and non-participants per format; effect size reported as rank-biserial correlation (*r*; positive = participants score higher); comparisons run only where both groups had $n \geq 30$, at $\alpha = .05$; unweighted, with a weighted (BRR) sensitivity check and a joint model with all formats entered together (school-clustered standard errors)

Key findings

Outcome: assessment self-efficacy (`TT4G27J`)

Valid responses: n = 1,179 of 1,271 (92.8%). Mean = 2.96, SD = 0.73, median = 3. Self-efficacy is **fairly high**: three-quarters of teachers answer “quite a bit” or “a lot”.

Response	n	%
1 – Not at all	31	2.6
2 – To some extent	247	20.9
3 – Quite a bit	642	54.5
4 – A lot	259	22.0
Total	**1,179**	**100.0**

Predictors: PD format participation

Percentages are of valid responses to each item.

Format	n valid	% valid	% participated
Lesson observation & reflection	1,217	95.8	69.0
Self-initiated learning	1,216	95.7	53.8
Education conferences	1,217	95.8	42.6
Teacher networks	1,213	95.4	40.7
Coaching / mentoring	1,214	95.5	35.9
Visits to other schools	1,214	95.5	23.8
Visits to business premises	1,215	95.6	20.3
Formal qualification programme	1,209	95.1	19.5

Format	n valid	% valid	% participated
Courses / workshops / seminars	0	0.0	—

Context item `TT4G21M` (PD covered analysis of student assessments): 38.0% Yes (n valid = 1,199); not analysed.

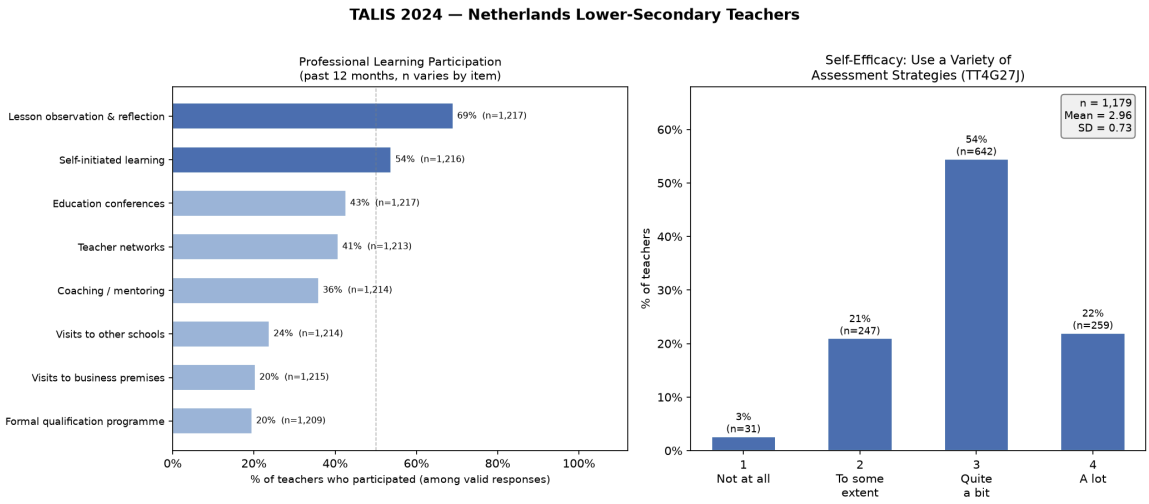


Figure 1: Descriptive overview

Eight formats had at least 30 teachers in both groups. **Three show small, statistically significant positive associations** with assessment self-efficacy; the other five do not:

Format	n Yes	n No	Mean Yes	Mean No	Diff [95% CI]	*p* (Mann-Whitney)	Effect (*r*)	Weighted diff (BRR)	Weighted *p*
Education conferences	497	672	3.06	2.89	+0.17 [0.09, 0.26]	< .001	+0.12	+0.196	< .001
Formal qualification programme	222	939	2.89	2.97	−0.08 [−0.19, 0.03]	.103	−0.06	−0.083	.079
Visits to other schools	276	889	3.04	2.93	+0.11 [0.01, 0.20]	.053	+0.07	+0.120	.076
Visits to business premises	238	929	3.03	2.94	+0.10 [−0.00, 0.19]	.106	+0.06	+0.066	.387
Lesson observation & reflection	802	367	2.98	2.91	+0.07 [−0.02, 0.16]	.170	+0.05	+0.073	.211
Coaching / mentoring	421	744	2.99	2.94	+0.05 [−0.03, 0.14]	.249	+0.04	+0.023	.612
Teacher networks	475	691	3.07	2.88	+0.20 [0.11, 0.28]	< .001	+0.14	+0.220	< .001
Self-initiated learning	629	539	3.04	2.86	+0.18 [0.10, 0.27]	< .001	+0.13	+0.198	< .001

Yes = participated in the format; *No* = did not. Means are on the 1–4 scale. Comparisons are complete-case: *n* excludes teachers with a missing answer on either variable. The 95% CI is a Welch *t* interval for the unweighted difference, so a CI that only just excludes zero (visits to other schools) can sit beside a Mann-Whitney *p* slightly above .05. Weighted columns use `TCHWGT` with 100 BRR replicate weights (Fay *k* = 0.5).

Courses/workshops (TT4G20A) has no valid Netherlands responses and was not tested. Participants in the three significant formats score about 0.17–0.20 points higher on the 4-point scale.

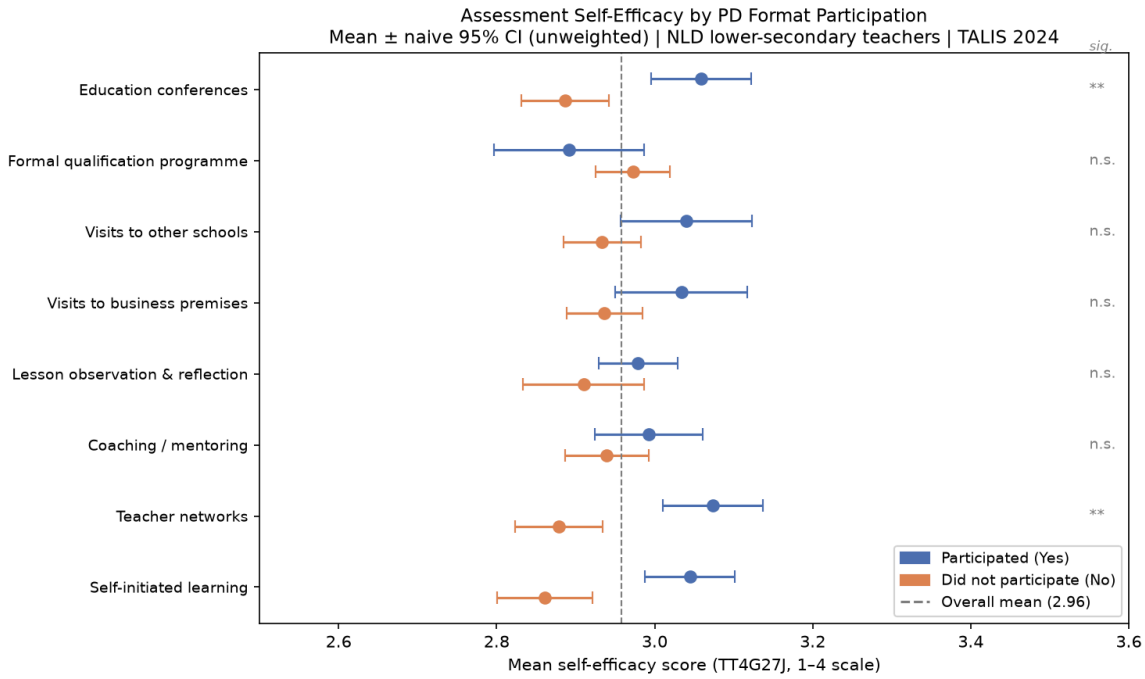


Figure 2: Group comparisons by format

Joint model (sensitivity check)

The tests above take one format at a time, but teachers who use one format tend to use others (pairwise correlations up to .31). To check that the three associations are not one effect counted three times, all eight formats were entered together in a single OLS model (school-clustered standard errors, unweighted, n = 1,135 complete cases).

Format	*b*	95% CI	*p*
Education conferences	+0.126	[0.035, 0.216]	.006
Formal qualification programme	–0.104	[–0.213, 0.005]	.061
Visits to other schools	+0.051	[–0.048, 0.151]	.312
Visits to business premises	+0.021	[–0.078, 0.120]	.673
Lesson observation & reflection	+0.028	[–0.057, 0.114]	.515
Coaching / mentoring	+0.004	[–0.075, 0.083]	.922
Teacher networks	+0.142	[0.067, 0.217]	< .001
Self-initiated learning	+0.121	[0.034, 0.207]	.006

Conferences, networks and self-initiated learning each remain positive and significant at .05 after adjusting for the other formats (coefficients of about 0.12–0.14 points). Against the Bonferroni threshold of .05 / 8 = .00625, networks and self-initiated learning pass, and conferences narrowly misses (p = .0064). An ordered logit, which treats the outcome as ordinal, gives the same three positive associations (odds ratios 1.42, 1.47 and 1.38; p = .004, < .001 and .009). Under that model the formal qualification programme turns weakly *negative* (OR = 0.74, p = .042), which was not significant in the OLS model (p =

.061) or the one-at-a-time test (p = .103), so it is treated as unstable and exploratory. Both models are sensitivity checks with 84 school clusters, no weights and no teacher or school covariates.

Interpretation

Teachers who attended education conferences, took part in teacher networks or engaged in self-initiated learning report somewhat higher assessment confidence in this sample. The differences are small ($r \approx$ 0.12–0.14) and the design cannot show whether the formats build confidence or whether more confident teachers choose them.

Two more theoretically proximate predictors warrant future analysis:

- **PD content** (`TT4G21M`): whether PD specifically covered student assessment analysis is a more direct test of the format–confidence link
- **Teacher and school covariates**: years of experience, gender and school type were not adjusted for, and a multilevel model would partition variance between teachers and schools

Limitations

- **Survey design**: the main tests are unweighted and ignore school clustering; a weighted sensitivity check (`TCHWGT` with BRR replicate weights) supports the same three associations, but results should still be read as sample-level associations
- **Dutch response rates**: the OECD Netherlands country note (7 October 2025) states that the Dutch sample did not meet TALIS participation-rate standards, with a higher risk of non-response bias; results describe the responding teachers and may not generalise to all Dutch lower-secondary teachers
- **Cross-sectional design**: causality cannot be inferred; teachers with higher baseline self-efficacy may self-select into more PD
- **Self-report bias**: both predictor and outcome are self-reported; self-efficacy scores may reflect personality disposition rather than genuine competence
- **Multiple comparisons**: eight tests with no formal adjustment (in the one-at-a-time tests the three significant results pass a Bonferroni threshold of .05 / 8 = .00625; in the joint model conferences narrowly misses it); formats are analysed one at a time, and the joint model adjusts for the other formats but for no teacher or school characteristics
- **Missing item**: `TT4G20A` (courses/workshops) has no valid Netherlands responses

Tools

Python: `pandas`, `numpy`, `scipy`, `matplotlib`, `pyreadstat` | Environment: Jupyter Notebook

References: OECD, *TALIS 2024 country note: Netherlands* (7 October 2025); OECD, TALIS 2024 technical documentation and international database (webfs.oecd.org/talis).

Data: TALIS 2024, OECD | Country: Netherlands (NLD2, lower-secondary) | Full analysis: `talis2024.ipynb`